46-929, Fall 2019: Homework #5

Due 5:30 PM, Wednesday, December 4, 2019

Rembember that you are to work **individually** on your homework assignments.

1. This exercise is based on page 399 in Ruppert and Matteson. The description they provide there states the following:

   > The data set `MacroVars.csv` contains three US macroeconomic indicators from Quarter 1 of 1959 to Quarter 4 of 1997: Real Gross Domestic Product (a measure of economic activity), Consumer Price Index (a measure of inflation), and Federal Funds Rate (a proxy for monetary policy). Each series has been transformed to stationary based on the procedures suggested by Stock and Watson (2005).

   This data set is provided on Canvas.

   (a) Construct the matrix of ACFs and CCFs for these three series. Comment on anything interesting that you find.

   (b) Fit a VAR(p) model. Use AIC to choose the value of p.

   (c) Consider the residuals from the above fit. Run an appropriate hypothesis test to determine if there is evidence of remaining serial correlation.

2. Consider the "R Lab" presented in Section 14.16.3 of Ruppert and Matteson. The needed data set is available on Canvas. Do Problems 7, 8, and 9. The necessary pages are attached.

Problem 4 *Now find an ARMA+GARCH model for the series* `diff.log.Tbill`*, which we will define as* `diff(log(Tbill))`*. Do you see any advantages of working with the differences of the logarithms of the T-bill rate, rather than with the difference of* `Tbill` *as was done earlier?*

14.16.2 The GARCH-in-Mean (GARCH-M) Model

A GARCH-in-Mean or *GARCH-M* model takes the form

$$\begin{aligned} Y_t &= \mu + \delta\sigma_t + a_t \\ a_t &= \sigma_t \epsilon_t \\ \sigma_t^2 &= \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2 \end{aligned}$$

in which $\epsilon_t \overset{iid}{\sim} (0,1)$. The GARCH-M model directly incorporates volatility as a regression variable. The parameter δ represents the *risk premium*, or reward for additional risk. Modern portfolio theory dictates that increased volatility leads to increased risk, requiring larger expected returns. The presence of volatility as a statistically significant predictor of returns is one of the primary contributors to serial correlation in historic return series. The data set `GPRO.csv()` contains the adjusted daily closing price of GoPro stock from June 26, 2014 to January 28, 2015.

Run the following `R` commands to fit a GARCH-M model to the GoPro stock returns.

```
library(rugarch)
GPRO = read.table("GPRO.csv")
garchm = ugarchspec(mean.model=list(armaOrder=c(0,0),
                                    archm=T,archpow=1),
                    variance.model=list(garchOrder=c(1,1)))
GPRO.garchm = ugarchfit(garchm, data=GPRO)
show(GPRO.garchm)
```

Problem 5 *Write out the fitted model. The parameter δ is equal to* `archm` *in the* `R` *output.*

Problem 6 *Test the one-sided hypothesis that $\delta > 0$ verses the alternative that $\delta = 0$. Is the risk premium significant?*

14.16.3 Fitting Multivariate GARCH Models

Run the following code to again load the data set `TbGdpPi.csv`, which has three variables: the 91-day T-bill rate, the log of real GDP, and the inflation rate. In this lab you will now use the first and third series after taking first differences.

```
TbGdpPi = read.csv("TbGdpPi.csv", header=TRUE)
TbPi.diff = ts(apply(TbGdpPi[,-2],2,diff), start=c(1955,2), freq=4)
plot(TbPi.diff)
acf(TbPi.diff^2)
source("SDAFE2.R")
mLjungBox(TbPi.diff^2, lag=8)
```

Problem 7 *Does the joint series exhibit conditional heteroskedasticity? Why?*

Now fit and plot a EWMA model with the following R commands.

```
EWMA.param = est.ewma(lambda.0=0.95, innov=TbPi.diff)
EWMA.param$lambda.hat
EWMA.Sigma=sigma.ewma(lambda=EWMA.param$lambda.hat,innov=TbPi.diff)
par(mfrow = c(2,2))
plot(ts(EWMA.Sigma[1,1,]^.5, start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(a) ", hat(sigma)["1,t"])))
plot(ts(EWMA.Sigma[1,2,], start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(b) ", hat(sigma)["12,t"])))
plot(ts(EWMA.Sigma[1,2,]/(sqrt(EWMA.Sigma[1,1,]* EWMA.Sigma[2,2,])),
        start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(c) ", hat(rho)["12,t"])))
plot(ts(EWMA.Sigma[2,2,]^.5, start = c(1955, 2), frequency = 4),
     type = 'l', xlab = "year", ylab = NULL,
     main = expression(paste("(d) ", hat(sigma)["2,t"])))
```

Problem 8 *What is the estimated persistence parameter λ?*

Now estimate standardized residuals and check whether they exhibit any conditional heteroskedasticity

```
n = dim(TbPi.diff)[1]
d = dim(TbPi.diff)[2]
stdResid.EWMA = matrix(0,n,d)
for(t in 1:n){
  stdResid.EWMA[t,] = TbPi.diff[t,] %*% matrix.sqrt.inv
  (EWMA.Sigma[,,t])
}
mLjungBox(stdResid.EWMA^2, lag=8)
```

Problem 9 *Based on the output of the Ljung-Box test for the squared standardized residuals, is the EWMA model adequate?*

Run the following command in R to determine whether the joint series are DOCs in volatility.

```
DOC.test(TbPi.diff^2, 8)
```